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CSC-207: Database Systems

Lab # 04: Exploring DQL Statements

Objectives

1. SQL foundation statements
2. DQL Statements

Data Query Language (DQL)

- DQL is used to fetch the data from the database.
- SELECT is the only and essential command widely and commonly used by all data analysts and scientists.

1. SELECT

The purpose of this statement is to get data from a database. Every query will begin with SELECT, followed by the names of columns you want to get from the table.

A particular column from the table can be selected with,

```
SELECT column_name FROM table_name;
```

And all the columns can be determined by using * as shown below,

```
SELECT * FROM table_name;
```

SQL foundation statements

1. USE Statement

To select a particular database to work with you issue the USE statement as follows:

```
USE database_name;
```

In this statement, following the USE keyword is the name of the database that you want to select.

2. SELECT Statement

The SELECT statement is used to select data from a database. The data returned is stored in a result table, called the result-set. A SELECT indicates that we are merely reading information, as opposed to modifying it. What we are selecting is identified by an expression or column list immediately following the SELECT. The FROM statement specifies the name of the table or tables from which we are getting our data.

When you want to select particular fields available in the table, use the following syntax:

```
SELECT * / column1, column2, ... FROM table_name;
```

Example:

```
SELECT FirstName, LastName FROM Employees;
```

Selects data of these two columns from the Employees table

When you want to select all the fields available in the table, use the following syntax:

```
SELECT * FROM table_name;
```

Example:

```
SELECT * FROM Employees;
```

Selects all the employees records from the database and displays its columns.

3. DISTINCT Statement

The SELECT DISTINCT statement is used to return only distinct (different) values.

Inside a table, a column often contains many duplicate values; and sometimes you only want to list the different (distinct) values.

```
Syntax: SELECT DISTINCT column1, column2, ... FROM table_name;
```

4. ORDER BY Keyword

The ORDER BY keyword is used to sort the result-set in ascending or descending order.

The ORDER BY keyword sorts the records in ascending order by default. To sort the records in

descending order, use the DESC keyword.

Syntax:

```
SELECT column_name(s) FROM table_name ORDER BY column1, column2, ... ASC|DESC
```

5. SELECT Limit Clause

The SELECT - LIMIT clause is used to specify the number of records to return.

The SELECT - Limit clause is useful on large tables with thousands of records. Returning a large number of records can impact performance.

MYSQL Syntax:

```
SELECT * FROM table_name LIMIT number;
```

The OFF SET value is also most often used together with the LIMIT keyword. The OFF SET value allows us to specify which row to start from retrieving data.

LIMIT with OFFSET Syntax:

```
SELECT * FROM table_name LIMIT OFFSET, number;
```

6. Functions with SELECT Statement

There are some functions that can be used in select statement.

Syntax: *SELECT function_name() FROM table_name;*

Functions are:

- UPPER() → converts text of column to uppercase: `SELECT UPPER(name) FROM employees;`
- LOWER() → converts text of column to lowercase: `SELECT LOWER(name) FROM employees;`
- LENGTH() → returns the number of characters in column: `SELECT LENGTH(name) FROM employees;`
- etc;

Aggregate functions used with GROUP BY:

- COUNT() → counts rows: `Select count(*) from employees;`
- SUM() → adds values: `Select sum(salary) from employees;`
- AVG() → calculates average: `Select avg(column_name) from employees;`
- MIN() → finds smallest value: `Select min(column_name) from table_name;`
- MAX() → finds largest value: `= Select max(column_name) from table_name;`

7. GROUP BY clause

The **GROUP BY** clause is used to **group rows that have the same values** in one or more columns. It is mainly used with **aggregate functions** to produce **summary results** for each group.

Why GROUP BY is used:

- To generate reports like “total students in each department”
- To find “average salary per department”
- To count “number of orders for each customer”

Important Note:

When using GROUP BY, any column in the SELECT list should either be:

- included in the GROUP BY clause, or
- used inside an aggregate function (like COUNT, SUM, etc.)

Syntax:

```
SELECT column_name, AGGREGATE_FUNCTION(column_name) FROM table_name GROUP BY column_name;
```

Example:

```
SELECT department_id, COUNT(*) FROM employees GROUP BY department_id;
```

8. Aliases

SQL aliases are used to give a table, or a column in a table, a temporary name. Aliases are often used to make column names more readable. An alias only exists for the duration of the query.

Alias Column Syntax:

```
SELECT column_name AS alias_name FROM table_name;
```

Alias Table Syntax:

```
SELECT column_name(s) FROM table_name AS alias_name;
```

Exercises (Class)

Add here all the tasks performed in lab.

Exercises (Weekly)

1. Write a query to display the names (first_name, last_name) using the alias name “First Name”, “Last Name”.
2. Write a query to get a unique department ID from the employee table.

3. Write a query to get all employee details from the employee table order by the first name, descending.
4. Write a query to get the employee ID, names (first_name, last_name), salary in ascending order of salary.
5. Write a query to get the total salaries payable to employees.
6. Write a query to get the maximum and minimum salary from the employee's table.
7. Write a query to get the average salary and number of employees in the employees' table.
8. Write a query to get the number of jobs available in the employee's table.
9. Write a query to get all first names from the employee's table in the upper case.
10. Write a query to select the first 10 records from a table.
11. Write a query to select the 3rd & 4th records of the employee's table.
12. Write a query to select 2nd last record of the employee's table.